

Identifying Risk Factor Regimes with Machine Learning: Implications for Tactical Asset Allocation

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1 Introduction

The distribution of asset prices exhibit time varying exposure to its underlying risk factors. This time variation in risk manifests either as a permanent structural break (i.e. secular decline in interest rates), or exhibits a persistent level of shock (i.e. equity market drawdown in Global Financial Crisis) which dissipates afterwards. These phenomena motivate investors to identify regimes for the underlying risk factors driving the dynamics of asset prices. Shifts in regimes have meaningful implication to the portfolio construction process, as it can cause significant deviations from investors' long term view of any asset. These deviations can either result in persistent sell-offs or may present opportunities over the short to medium term. Thus, modelling regimes of a risk factor and overlaying the forecasts on top a strategic asset allocation process (by tilting the risk allocation of a portfolio) can potentially aid in providing an attractive risk-return profile over the long run.

In practice, traditional methods of regime detection rely on generative approaches based on markov models (Hidden Markov Model), trend filtering methods (i.e. Moving Average, L1 Trend Filter, and Hodrick Prescott Filter etc.) and, simple rule based approaches (i.e. threshold based approaches based on intuition of the underlying characteristics) etc. There are potentially two main caveats to these approaches.

First, these approaches only tend to rely on either the asset in question or only a subset of underlying features which are deemed to be driving the underlying dynamics. However, markets are adaptive and far more complex, thus reliance on a few subset of features (either used directly or through dimension reduction techniques) may lead to serious inaccuracy in prediction during out of sample, under the case where those features may not necessarily be relevant anymore. Second, the traditional models used to predict regimes are mostly data driven. The data driven approaches accompanied by the parametric framework is dependent upon a set of assumptions related to the underlying distribution and state process, which may be less realistic in nature. Given that one has to explicitly state the linear, non-linear and interaction effects of the features, which leads to a large set of features being used and likely not favorable for these parametric approaches. Furthermore, there are interpretation issues, since the data driven nature of the methodology doesn't necessarily provide an intuitive approach to understanding and interpreting the underlying drivers other than what is specified by the system.

We adopt a supervised machine learning (ML) based approach to modelling regimes. ML approaches are beneficial for this exercise since it enables to not only work with a large set of features but also tackle overfitting issues which arise in most traditional models. In ML, the overfitting issues are mitigated through the tuning of hyper parameters conducted during the calibration process. Furthermore, inspired by cooperative game theory, using a novel method called 'Shapley Values' we also provide intuitive interpretation by analysing the prediction

contribution for each feature used in the model. The shapley values answer the concerns raised related to the 'Black Box' nature of most ML algorithms.

Our methodology is focused on modelling the regimes for the Equity Factor. The reason behind choosing the Equity Factor, as we will define it later, is mostly driven by the favor ability this particular source of risk has received by institutionals and small alternatives alike. Most traditional assets, systematic or discretionary process tend to have some or significant exposure in equity risk, and thus managing such exposure is crucial for portfolio construction. However, this analysis can also be replicated for other traditional factors, like interest rates, short volatility, and inflation etc.; and also for alternative risk factors like value, carry and, momentum etc, depending upon an investor's composition of the portfolio and which sources of risk is he/she most exposed to.

We first tag historical periods into two distinctive regimes, "Crash" and "Normal". Crash periods are essentially defined as regimes of persistent sell offs below a given threshold, high market volatility and convergence of systematic correlation across assets. Next, we incorporate our methodology to predict these regimes ex-ante. We present several performance metrics to assess the robustness, and classification accuracy, alongside performance measure by actually applying model outputs to a simple trading strategy. Finally, we also conduct benchmark analysis by showing that our method outperforms traditional VIX based, trend following rules, and markov models by minimizing false positives and improving risk adjusted return of a simple trading strategy. We also show a few applications of how the predicted probabilities can be used as position size for strategic tilts and also be incorporated in investor's view of risk. Finally, we tackle the issue with model interpretability with 'Shapley Values'.

2 Why Model Factor Regimes?

To understand how modelling regimes and applying a regime based portfolio construction approach can improve upon the pre-existing asset allocation framework, we refer to López de Prado (2018). Consider a primary asset/discretionary or systematic process/factor (i.e. long only equities in our example) which produces T IID bets each year (i.e. let's assume it to be the number of trading days in our case), with R_t being the excess return of the factor. Now, let's assume there are two distinct regimes with the positive payoff r_+ with probability p and the negative payoff r_- with probability $1 - p$, with $r_+ > r_-$. The expected return of this factor can be written as,

$$\begin{aligned}\mathbb{E}[R_t] &= pr_+ + (1 - p)r_- \\ &= (r_+ - r_-)p - r_-\end{aligned}\tag{1}$$

The variance of the factor can be written as,

$$\begin{aligned}\mathbb{V}[R_t] &= \mathbb{E}[R_t^2] - \mathbb{E}[R_t]^2 \\ &= (r_+ - r_-)^2 p(1 - p)\end{aligned}\tag{2}$$

The annualized Sharpe ratio of the factor can be written as,

$$S.R. = \frac{(r_+ - r_-)p - r_-}{(r_+ - r_-)\sqrt{p(1 - p)}}\sqrt{T}\tag{3}$$

If the factor is generally unprofitable, i.e. the positive payoff is less than the absolute values of negative payoffs, $r_+ < -r_-$, it may still be possible to increase the Sharpe ratio of the factor by improving the probability with the expense of T , meaning a large number of bets T can easily improve the sharpe ratio with a small increase in the probability and vice versa. In order to understand improving p leads to better risk adjusted returns, let us

understand the key components of a confusion matrix. Table 1 shows an illustration of the confusion matrix.

Table 1: Confusion Matrix

		$\hat{y}$	
		1	0
y	1	True Positive (TP)	False Negative (FN)
	0	False Positive (FP)	True Negative (TN)

$\hat{y}$ is the predicted outcome of our asset (i.e. a portfolio manager's expected return of equities) and y is the realized outcome of the asset. The elements inside the matrix are self-explanatory. Next, we define two important metrics, Precision and Recall.

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

Precision tells us out of all the predicted outcomes, what percentage of the predictions were profitable. On the other hand, Recall denotes what percentage of profitable outcomes are predicted accurately by our pre-existing expected return framework. An investment factor where the magnitude of returns are larger on the false positive outcomes as opposed to the true positive outcomes (typically the case in most traditional convergent asset classes with negatively skewed distributions) can lead to an overall unprofitable outcome. This is why it is important to quantify the likelihood of a regime shift and overlay with our primary prediction of the factor (which is essentially "always have long exposure in equity risk") to ensure we are not getting caught up in false positive outcomes with large market moves. The overall profitability can be improved by giving up some of the recall to improve its precision, thus lowering false positives and improving true negatives. Supervised classification algorithms can aid us in estimating probabilities of factor regimes and use the probabilities as a risk management overlay to our existing portfolio construction process.

3 Equity Factor

I define the Equity factor as a basket of seven equity indices covering major global regions and represents the factor exposure to global growth risk. The composition of the basket and key statistics are as follow,

Table 2: Global Equity Factor

Region	Weight
US	50%
CA	8.3%
UK	8.3%
EU	8.3%
AUS	8.3%
JP	8.3%
EM	8.3%

Table 2 continued from previous page

Region	Weight
Annualized Statistics	
Arithmetic Mean	10.09%
Compound Mean	9.55%
Volatility	13.90%
Sharpe Ratio	0.37

4 Tagging Equity Factor Regimes

In order to tag the crash regime, I start off with the daily equity factor total return series constructed from the previous section. First, I scale the time series to 10% target volatility using an expanding window volatility estimate. That is,

$$r_{EQ,t}^{scaled} = r_{EQ,t}^{raw} * \frac{10\%}{\sigma_{[t_0:t]} \sqrt{252}} \quad (6)$$

$\sigma_{[t_0:t]}$ is the expanding window daily volatility estimate using all information upto time t and $r_{EQ,t}^{raw}$ is the raw equity factor return. Next, I cumulate the time series to construct a curve and compute a measure for drawdown at each point in time as,

$$DD_t = \frac{P_t}{\max\{P_i\}_{i=t_0}^t} - 1 \quad (7)$$

Here, P_t is the cumulative scaled return index. Using the underwater curve, I consider all possible periods in time where we have seen drawdown in excess of 10%, which is roughly 1 vol point move for time series with 10% conditional volatility. To clarify the calculations a bit further, I iterate over all possible peak to trough in the data and assess whether the cumulative return was less than 10%. Furthermore, not all trough to recovery periods are smooth and can be subjected to choppy movements along the way while witnessing a local drawdown even satisfying our threshold. As such, for any peak to trough which satisfy the threshold, I also evaluate whether its trough to recovery period demonstrated any local drawdowns and tag those periods accordingly.

To illustrate this, let's look at the underwater curve as shown below.

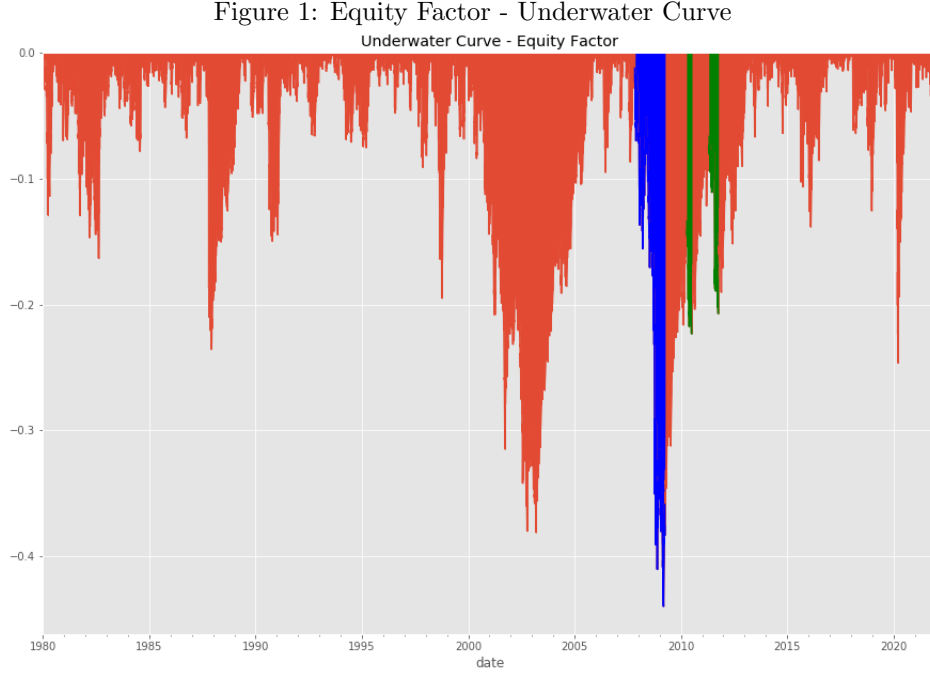
The longest peak to trough during GFC was from the November of 2007 up until the March of 2009 (colored blue). However, it took until the February 2013 to recover from the trough, around 4 years. This was partially to few local drawdown periods witnessed around this, notably the 2010 Flash Crash and also the 2011 Euro Sovereign Debt Crisis (colored green). These local crash regimes are also crucial to be taken into account when trying to estimate the crash probability for equities, especially since these events are very equity centric and do not overlap with recessionary periods. Thus, I construct a local underwater curve for those recovery periods and tag crash regimes based on the threshold assigned. The tag is 1 if the period breaches the -10% threshold and 0 otherwise.

The regime at time t , denoted as y_t is computed as follows,

$$y_t = \mathbf{1}_{[t \in pt_\tau \text{ or } t \in tr_\tau]} \quad (8)$$

$$pt_\tau = [\tau_{tagged \text{ peak}} : \tau_{tagged \text{ trough}}] \quad (9)$$

$$tr_\tau = [\tau_{tagged \text{ trough}} : \tau_{tagged \text{ recovery}}] \mathbf{1}_{[t \in pt_\tau]} \quad (10)$$



Where, pt_τ are peak to trough periods satisfying our -10% threshold and tr_τ are trough to recovery periods which follows a tagged peak to trough periods and where the local drawdowns also satisfy the -10% threshold.

5 General Framework

In its most general form we define the probability of a crash regime as,

$$P(y_{t+\tau} = 1 \mid X_t) = f(X_t) \quad (11)$$

τ is the forecast horizon, in this case we use 1-month or 21 trading days, $f(X_t)$ is a general functional form of the machine learning algorithms with X_t being a vector of features/variables at time t .

Also note from this point forth, I will use 'features' and 'variables' interchangeably.

6 Data Description

For the set of features, I initially focus on using real-time variables. This leads us to mostly rely on market variables and surprise indices. As for the data source, I mostly rely on Bloomberg, and DataStream. The initial set of variables using realtime/market related variables can be divided into 19 segments,

Table 3:

List of Initial Variable Categories
Equity Total Return Indices for major DMs and EM
Bond total return indices for major DMs
FX Forwards

Table 3 continued from previous page

List of Initial Variable Categories
Commodity Total Return Indices by category
CBOE Volatility indices
S&P 500 Volatility Surface for three maturities (1m, 3m, and 6m)
Term Structure Information from VIX Futures
Historical Index Implied Volatility for key Indices
USD Swaption Volatility
FX Implied Volatility
High Yield spreads for three major regions
Economic Surprise Indices for major DMs
Inflation Surprise Indices for major DMs
CFTC positioning Indices for few assets
US Option Volume/Positioning variables
Index Breadth for key Indices
EPS revision for key indices
Nominal Yields for major DMs
Breakeven Yields for major DMs
Sovereign CDS
Central Bank Rate Expectation from Futures Market
Key Liquidity Risk Measures
Sector Level Total Return Indices for Major DMs

The IDs of the equity sector data from DataStream and A detail outline of the data sourced from Bloomberg are listed in the tables below,

Table 4: DataStream IDs for the Regional Sector Level Data

US	EU	JP	UK	CA	AUS	EM
TECNOUS(RI)	TOTMKEM	TOTMKJP	TOTMKUK	TOTMKCN	TOTMKAU	TOTMKEK
TELCMUS(RI)	TECNOEM	TECNOJP	TECNOUK	TECNOCN	TECNOAU	TECNOEK
HLTHCUS(RI)	HLTHCEM	HLTHCJP	HLTHCUK	HLTHCCN	HLTHCAU	HLTHCEK
FINANUS(RI)	CODISEM	CODISJP	CODISUK	CODISCN	CODISAU	CODISEK
RLESTUS(RI)	TELCMEM	TELCMJP	TELCMUK	TELCMCN	TELCMAU	TELCMEK
COSTPUS(RI)	FINANEM	FINANJP	FINANUK	FINANCN	FINANAU	FINANEK
INDUSUS(RI)	INDUSEM	INDUSJP	INDUSUK	INDUSCN	INDUSAU	INDUSEK
BMATRUS(RI)	COSTPEM	COSTPJP	COSTPUK	COSTPCN	COSTPAU	COSTPEK
ENERGYUS(RI)	UTILSEM	UTILSJP	UTILSUK	UTILSCN	UTILSAU	UTILSEK
UTILSUS(RI)	RLESTEM	RLESTJP	RLESTUK	RLESTCN	RLESTAU	RLESTEK
CODISUS(RI)	BMATREM	BMATRJP	BMATRUUK	BMATRCN	BMATRAU	BMATREK
	ENERGYEM	ENERGYJP	ENERGYUK	ENERGYCN	ENERGYAU	ENERGYEK

Figure 2: Tagged Equity Factor Regimes
Tagged Crisis Regimes - Target Variable

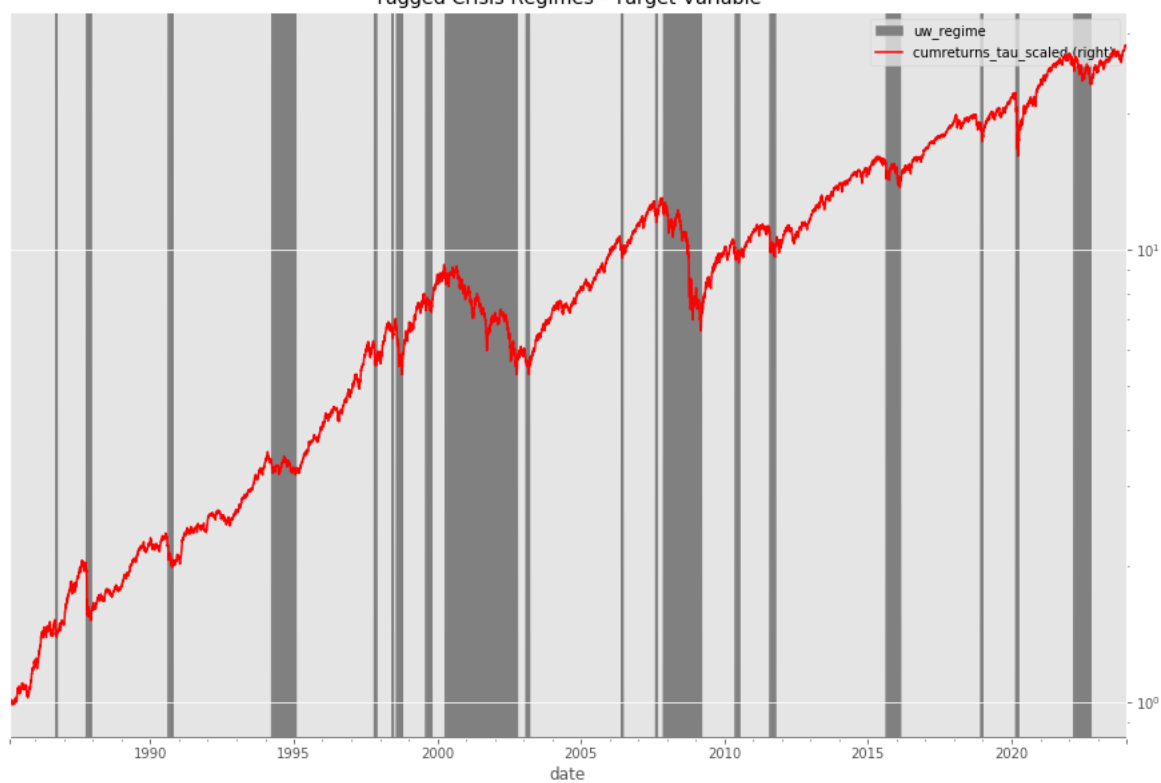


Table 5: Description of Initial Features Sourced from Bloomberg

Category	ID	Ticker	Field	tcode
eq_prices	spx_totalr	SPX Index	TOT_RETURN_INDEX_GROSS_DVDS	1
eq_prices	sptsx_totalr	SPTSX Index	TOT_RETURN_INDEX_GROSS_DVDS	1
eq_prices	tpx_totalr	TPX500 Index	TOT_RETURN_INDEX_GROSS_DVDS	1
eq_prices	sxxe_totalr	SXXE Index	TOT_RETURN_INDEX_GROSS_DVDS	1
eq_prices	as51_totalr	AS51 Index	TOT_RETURN_INDEX_GROSS_DVDS	1
eq_prices	mxef_totalr	MXEF Index	TOT_RETURN_INDEX_GROSS_DVDS	1
bond_prices	ml_10y_us_trsry_futres	MLT1US10 Index	PX_LAST	1
bond_prices	sp_10y_ca_trsry_futres	SPCACGTR Index	PX_LAST	1
bond_prices	sp_10y_jp_trsry_futres	SPJGBTR Index	PX_LAST	1
bond_prices	sp_near_eu_trsry_futres	SPEUBDTR Index	PX_LAST	1
bond_prices	sp_10y_au_trsry_futres	SPAU10DT Index	PX_LAST	1
fx	cad_1m_fwd	CAD1M BGN Curncy	PX_LAST	2
fx	cad_6m_fwd	CAD6M BGN Curncy	PX_LAST	2
fx	jpy_1m_fwd	JPY1M BGN Curncy	PX_LAST	2
fx	jpy_6m_fwd	JPY6M BGN Curncy	PX_LAST	2
fx	eur_1m_fwd	EUR1M BGN Curncy	PX_LAST	2
fx	eur_6m_fwd	EUR6M BGN Curncy	PX_LAST	2
fx	aud_1m_fwd	AUD1M BGN Curncy	PX_LAST	2
fx	aud_6m_fwd	AUD6M BGN Curncy	PX_LAST	2
fx	em_fxy	FXCTEM8 Index	PX_LAST	1
fx	dxy_index	FXCTEM8 Index	PX_LAST	2
comms	gsci.energy	SPGSENTR Index	PX_LAST	1
comms	gsci.ind.metals	SPGSINTR Index	PX_LAST	1
comms	gsci_pre.metals	SPGSPMTR Index	PX_LAST	1

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
comms	gsci_ag	SPGSAGTR Index	PX_LAST	1
comms	gsci_livestock	SPGSALTR Index	PX_LAST	1
comms	gold	GC1 A:03.0_R Comdtty	PX_LAST	1
comms	silver	SI1 A:03.0_R Comdtty	PX_LAST	1
vol_prices	cboe_sp_5pct_put	PPUT Index	PX_LAST	1
vol_prices	cboe_sp_short_put	PUT Index	PX_LAST	1
vol_prices	cboe_sp_buywrite	BXM Index	PX_LAST	1
vol_indices	vix_index	VIX Index	PX_LAST	0
vol_indices	vix3m_index	VIX3M Index	PX_LAST	0
vol_indices	vnky_index	VNKY index	PX_LAST	0
vol_indices	v1x_index	V1X index	PX_LAST	0
vol_indices	vstoxx_index	V2X index	PX_LAST	0
vol_indices	move	MOVE Index	PX_LAST	0
vol_indices	vdax	V1X Index	PX_LAST	0
vol_indices	vxeem	VXEEM Index	PX_LAST	0
vol_indices	ivolgold	IVOLGOLD Index	PX_LAST	0
vol_indices	cvix	CVIX Index	PX_LAST	0
vol_indices	ivolcrud	IVOLCRUD Index	PX_LAST	0
spx_vol_surface	spx_3m_90_imp_vol	SPX Index	3MTH_IMPVOL_90.0%MNY_DF	0
spx_vol_surface	spx_3m_100_imp_vol	SPX Index	3MTH_IMPVOL_100.0%MNY_DF	0
spx_vol_surface	spx_3m_110_imp_vol	SPX Index	3MTH_IMPVOL_110.0%MNY_DF	0
sx5e_vol_surface	sx5e_3m_90_imp_vol	SX5E Index	3MTH_IMPVOL_90.0%MNY_DF	0
sx5e_vol_surface	sx5e_3m_100_imp_vol	SX5E Index	3MTH_IMPVOL_90.0%MNY_DF	0
sx5e_vol_surface	sx5e_3m_110_imp_vol	SX5E Index	3MTH_IMPVOL_110.0%MNY_DF	0
nky_vol_surface	nky_3m_90_imp_vol	NKY Index	3MTH_IMPVOL_90.0%MNY_DF	0
nky_vol_surface	nky_3m_100_imp_vol	NKY Index	3MTH_IMPVOL_100.0%MNY_DF	0
nky_vol_surface	nky_3m_110_imp_vol	NKY Index	3MTH_IMPVOL_110.0%MNY_DF	0
spx_vol_surface	spx_30d_100_imp_vol	SPX Index	30DAY_IMPVOL_100.0%MNY_DF	0
sx5e_vol_surface	sx5e_30d_100_imp_vol	SX5E Index	30DAY_IMPVOL_100.0%MNY_DF	0
nky_vol_surface	nky_30d_100_imp_vol	NKY Index	30DAY_IMPVOL_100.0%MNY_DF	0
historical_index_implied_vol	spx_index_hist_put_imp_vol	SPX Index	HIST_PUT_IMP_VOL	0
historical_index_implied_vol	cac_index_hist_put_imp_vol	CAC Index	HIST_PUT_IMP_VOL	0
historical_index_implied_vol	ukx_index_hist_put_imp_vol	UKX Index	HIST_PUT_IMP_VOL	0
historical_index_implied_vol	nky_index_hist_put_imp_vol	NKY Index	HIST_PUT_IMP_VOL	0
swaption_vol	usd_swap1m2y_imp_vol	USSV0A2 Curncy	PX_LAST	0

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
swaption_vol	usd_swap3m2y_imp_vol	USSV0C2 Curncy	PX_LAST	0
swaption_vol	usd_swap6m2y_imp_vol	USSV0F2 Curncy	PX_LAST	0
swaption_vol	usd_swap1m10y_imp_vol	USSV0A10 Curncy	PX_LAST	0
swaption_vol	usd_swap3m10y_imp_vol	USSV0C10 Curncy	PX_LAST	0
swaption_vol	usd_swap6m10y_imp_vol	USSV0F10 Curncy	PX_LAST	0
swaption_vol	eu_swap1m2y_imp_vol	EUSV0A2 BGN Curncy	PX_LAST	0
swaption_vol	eu_swap3m2y_imp_vol	EUSV0C2 BGN Curncy	PX_LAST	0
swaption_vol	eu_swap6m2y_imp_vol	EUSV0F2 BGN Curncy	PX_LAST	0
swaption_vol	eu_swap1m10y_imp_vol	EUSV0A10 BGN Curncy	PX_LAST	0
swaption_vol	eu_swap3m10y_imp_vol	EUSV0C10 BGN Curncy	PX_LAST	0
swaption_vol	eu_swap6m10y_imp_vol	EUSV0F10 BGN Curncy	PX_LAST	0
swaption_vol	jpy_swap1m2y_imp_vol	JPNEA2 BGN Curncy	PX_LAST	0
swaption_vol	jpy_swap3m2y_imp_vol	JPNEC2 BGN Curncy	PX_LAST	0
swaption_vol	jpy_swap6m2y_imp_vol	JPNEF2 BGN Curncy	PX_LAST	0
swaption_vol	jpy_swap1m10y_imp_vol	JPNEA10 BGN Curncy	PX_LAST	0
swaption_vol	jpy_swap3m10y_imp_vol	JPNEC10 BGN Curncy	PX_LAST	0
swaption_vol	jpy_swap6m10y_imp_vol	JPNEF10 BGN Curncy	PX_LAST	0
fx_risk_reversal	fxsk_usdjpy	USDJPY25R3M BGN Curncy	PX_LAST	0
fx_risk_reversal	fxsk_eurjpy	EURJPY25R3M BGN Curncy	PX_LAST	0
fx_risk_reversal	fxsk_audjpy	AUDJPY25R3M BGN Curncy	PX_LAST	0
fx_implied_vol	usd-cad_opt_vol_1m	USDCADV1M BGN Curncy	PX_LAST	0
fx_implied_vol	usd-cad_opt_vol_3m	USDCADV3M BGN Curncy	PX_LAST	0

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
fx IMPLIED VOL	usd-cad_opt_vol_6m	USDCADV6M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	usd-jpy_opt_vol_1m	USDJPYV1M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	usd-jpy_opt_vol_3m	USDJPYV3M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	usd-jpy_opt_vol_6m	USDJPYV6M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	eur-usd_opt_vol_1m	EURUSDV1M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	eur-usd_opt_vol_3m	EURUSDV3M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	eur-usd_opt_vol_6m	EURUSDV6M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	aud-usd_opt_vol_1m	AUDUSDV1M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	aud-usd_opt_vol_3m	AUDUSDV3M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	aud-usd_opt_vol_6m	AUDUSDV6M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	usd-cny_opt_vol_1m	USDCNYV1M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	usd-cny_opt_vol_3m	USDCNYV3M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	usd-cny_opt_vol_6m	USDCNYV6M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	xau-usd_opt_vol_1m	XAUUSDV1M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	xau-usd_opt_vol_3m	XAUUSDV3M BGN Curncy	PX_LAST	0
fx IMPLIED VOL	xau-usd_opt_vol_6m	XAUUSDV6M BGN Curncy	PX_LAST	0
economic surprises	bbg_ecosurprise_us	ECSURPUS Index	PX_LAST	0
economic surprises	bbg_ecosurprise_em	ECSURPEA Index	PX_LAST	0
economic surprises	bbg_ecosurprise_paneu	ECSURPGB Index	PX_LAST	0
economic surprises	citi_econ_surprise_us	CESIUSD In- dex	PX_LAST	0
economic surprises	citi_econ_surprise_ca	CESICAD In- dex	PX_LAST	0
economic surprises	citi_econ_surprise_uk	CESIGBP In- dex	PX_LAST	0

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
economic_surprises	citi_econ_surprise_eu	CESIEUR In- dex	PX_LAST	0
economic_surprises	citi_econ_surprise_au	CESIAUD In- dex	PX_LAST	0
economic_surprises	citi_econ_surprise_jp	CESIJPY In- dex	PX_LAST	0
economic_surprises	citi_econ_surprise_em	CESIEM In- dex	PX_LAST	0
inflation_surprises	citi_inf_surprise_us	CSIIUSD In- dex	PX_LAST	0
inflation_surprises	citi_inf_surprise_ca	CSII CAD In- dex	PX_LAST	0
inflation_surprises	citi_inf_surprise_uk	CSII GBP In- dex	PX_LAST	0
inflation_surprises	citi_inf_surprise_eu	CSII EUR In- dex	PX_LAST	0
inflation_surprises	citi_inf_surprise_au	CSII AUD In- dex	PX_LAST	0
inflation_surprises	citi_inf_surprise_jp	CSII JPY In- dex	PX_LAST	0
inflation_surprises	citi_inf_surprise_em	CSII EM Index	PX_LAST	0
cftc_cot_positions	spx_net_pos	IMM0ENCN Index	PX_LAST	0
cftc_cot_positions	spx_open_interest	IMM0EOIN Index	PX_LAST	0
cftc_cot_positions	ndx_net_pos	IMM3NCON Index	PX_LAST	0
cftc_cot_positions	ndx_open_interest	IMM3NOIN Index	PX_LAST	0
cftc_cot_positions	nyce_net_pos	NYC2UNCN Index	PX_LAST	0
cftc_cot_positions	nyce_open_interest	NYC2UOIN Index	PX_LAST	0
cftc_cot_positions	oil_net_pos	NYM1CNCN Index	PX_LAST	0
cftc_cot_positions	oil_open_interest	NYM1COIN Index	PX_LAST	0
cftc_cot_positions	us2y_net_pos	CBT42NCN Index	PX_LAST	0
cftc_cot_positions	us2y_open_interest	CBT42OIN Index	PX_LAST	0
cftc_cot_positions	us10y_net_pos	CBT4TNCN Index	PX_LAST	0

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
cftc_cot_positions	us10y_open_interest	CBT4TOIN Index	PX_LAST	0
us_option_volumes	us_total_option_call_volumes	OPCVTOTC Index	PX_LAST	3
us_option_volumes	us_total_option_put_volumes	OPCVTOTP Index	PX_LAST	3
us_option_volumes	us_put_call_ratio_composite	OPCVTPCR Index	PX_LAST	0
us_option_volumes	us_put_call_oi_ratio_composite	SPXPCOI U Index	PX_LAST	0
index_breadth	spx_pct_above200D	SPX Index	PCT_MEMB.ABOVE.MOV_AVG_200D	0
index_breadth	ndx_pct_above200D	CCMP Index	PCT_MEMB.ABOVE.MOV_AVG_200D	0
index_breadth	sptsx_pct_above200D	SPTSX Index	PCT_MEMB.ABOVE.MOV_AVG_200D	0
index_breadth	sx5e_pct_above200D	SX5E Index	PCT_MEMB.ABOVE.MOV_AVG_200D	0
index_breadth	ukx_pct_above200D	UKX Index	PCT_MEMB.ABOVE.MOV_AVG_200D	0
index_breadth	cac_pct_above200D	CAC index	PCT_MEMB.ABOVE.MOV_AVG_200D	0
index_breadth	nky_pct_above200D	NKY Index	PCT_MEMB.ABOVE.MOV_AVG_200D	0
index_breadth	hsi_pct_above200D	HSI Index	PCT_MEMB.ABOVE.MOV_AVG_200D	0
eps_revisions	citi_earning_revi_us	CGERUSA1 Index	PX_LAST	0
eps_revisions	citi_earning_revi_uk	CGERUKUK Index	PX_LAST	0
eps_revisions	citi_earning_revi_eu	CGEREUXU Index	PX_LAST	0
eps_revisions	citi_earning_revi_jp	CGERJAPN Index	PX_LAST	0
eps_revisions	citi_earning_revi_apac	CGERAPXJ Index	PX_LAST	0
eps_revisions	citi_earning_revi_em	CGERGLEM Index	PX_LAST	0
nominal_yields	us10y	USGG10YR Index	PX_LAST	4
nominal_yields	us3y	USGG3YR Index	PX_LAST	4
nominal_yields	us3m	USGG3M In- dex	PX_LAST	4
nominal_yields	ca10y	GCAN10YR Index	PX_LAST	4
nominal_yields	ca3y	GCAN3YR Index	PX_LAST	4
nominal_yields	ca3m	GCAN3M In- dex	PX_LAST	4
nominal_yields	de10y	GTEUR10Y GOVT	PX_LAST	4

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
nominal_yields	de3y	GTEUR3Y GOVT	PX_LAST	4
nominal_yields	de3m	GTEUR3M GOVT	PX_LAST	4
nominal_yields	uk10y	GUKG10 In- dex	PX_LAST	4
nominal_yields	uk3y	GUKG3 Index	PX_LAST	4
nominal_yields	uk3m	BP0003M In- dex	PX_LAST	4
nominal_yields	au10y	GACGB10 Index	PX_LAST	4
nominal_yields	au3y	GACGB3 In- dex	PX_LAST	4
nominal_yields	au3m	BBSW3M In- dex	PX_LAST	4
nominal_yields	jp10y	GJGB10 Index	PX_LAST	4
nominal_yields	jp3y	GJGB3 Index	PX_LAST	4
nominal_yields	jp3m	GJGB3M In- dex	PX_LAST	4
breakevens.ilbs	us_be_5	USGGBE05 Index	PX_LAST	4
breakevens.ilbs	us_be_10	USGGBE10 Index	PX_LAST	4
breakevens.ilbs	uk_gilts5y	GUKG5 Index	PX_LAST	4
breakevens.ilbs	uk_be_10	UKGGBE10 Index	PX_LAST	4
corp_credit	finsub_credit	ITRXEUE CBGT Index	PX_LAST	3
corp_credit	itraxx_japan	ITRXAJE CBGT Index	PX_LAST	3
corp_credit	itraxx_xover	ITRXEXE CBGT Index	PX_LAST	3
corp_credit	us_hy	IBOXHYSE CBIL INDEX	PX_LAST	3
corp_credit	us_avg_oas	LUACOAS Index	PX_LAST	3
corp_credit	us_high_yield_oas	LF98TRUU Index	INDEX_OAS.TSY_BP	0
corp_credit	eu_high_yield_oas	LP01TREU Index	INDEX_OAS.TSY_BP	0

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
corp_credit	em_usd_high_yield_oas	I31732US In- dex	INDEX_OAS.TSY_BP	0
us_real_rates	usd_10y_real_rate	GTII10 Govt	PX_LAST	4
us_real_rates	usd_5y_real_rate	GTII5 Govt	PX_LAST	4
us_real_rates	usd_20y_real_rate	GTII20 Govt	PX_LAST	4
us_real_rates	usd_30y_real_rate	GTII30 Govt	PX_LAST	4
tightening_expectations	fed_qt_expectations	USSN015 Curncy	PX_LAST	0
tightening_expectations	boe_qt_expectations	BPNE15 Curncy	PX_LAST	0
tightening_expectations	ecb_qt_expectations	EUNE15 Curncy	PX_LAST	0
tightening_expectations	boj_qt_expectations	JPNE15 Curncy	PX_LAST	0
tightening_expectations	hkma_qt_expectations	HDSN015 BBIR Curncy	PX_LAST	0
tightening_expectations	bom_qt_expectations	MPSN015 BBIR Curncy	PX_LAST	0
tightening_expectations	snb_qt_expectations	SFSN015 BBIR Curncy	PX_LAST	0
liquidity	euribor	EUR003M In- dex	PX_LAST	0
liquidity	eonia	EONIA Index	PX_LAST	0
liquidity	eur_3m_swap	EESWEC BGN Curncy	PX_LAST	0
liquidity	ted_spread	.TEDSP G Index	PX_LAST	0
liquidity	japan_libor_ois_spread	.JPLIBOIS G Index	PX_LAST	0
liquidity	us_libor_ois_spread	BICLOISS In- dex	PX_LAST	0
liquidity	us_sofr_ois_spread	.SOFROISS G Index	PX_LAST	0
country_cds	france_5y_cds_cbgt	CFRTR1U5 CBGT Curncy	PX_LAST	0
country_cds	france_5y_cds_helc	CFRTR1U5 HELC Curncy	PX_LAST	0
country_cds	italy_5y_cds	CITLY1U5 HELC Curncy	PX_LAST	0

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
country_cds	china_5y_cds	CCHIN1U5 CBGT Curncy	PX_LAST	0
country_cds	brazil_5y_cds	CBRZ1U5 CBGT Curncy	PX_LAST	0
country_cds	mexico_5y_cds	CMEX1U5 CBGT Curncy	PX_LAST	0
country_cds	south_africa_5y_cds	CSOAF1U5 CBGT Curncy	PX_LAST	0
country_cds	em_cds	IBOXUMSE CBGT Curncy	PX_LAST	0
country_cds	us_5y_cds	US CDS EUR SR 5Y D14 PRXY Corp	PX_LAST	0
country_cds	uk_5y_cds	UK CDS USD SR 5Y D14 PRXY Corp	PX_LAST	0
country_cds	korea_5y_cds	CKREA1U5 Curncy	PX_LAST	0
country_cds	hong_kong_5y_cds	CHKS1U5 PRXY Curncy	PX_LAST	0
country_cds	india_5y_cds	CIGB1U5 PRXY Curncy	PX_LAST	0
country_cds	german_5y_cds_cbgt	CDBR1U5 CBGT Curncy	PX_LAST	0
country_cds	german_5y_cds_helc	CDBR1U5 HELC Curncy	PX_LAST	0
country_cds	australia_5y_cds	CT855561 PRXY Curncy	PX_LAST	0
country_cds	japan_5y_cds_cbgt	CJGB1U5 CBGT Curncy	PX_LAST	0
country_cds	japan_5y_cds_prxy	CJGB1U5 PRXY Curncy	PX_LAST	0

Table 5 continued from previous page

Category	ID	Ticker	Field	tcode
rate_expectation	japan_rate_expectation_6m	YE6 N:01.0.D Comdty	PX_LAST	3
rate_expectation	japan_rate_expectation_2m	YE2 N:01.0.D Comdty	PX_LAST	3
rate_expectation	euro_rate_expectation_6m	ER6 N:01.0.D Comdty	PX_LAST	3
rate_expectation	euro_rate_expectation_2m	ER2 N:01.0.D Comdty	PX_LAST	3
rate_expectation	us_rate_expectation_6m	FF6 N:01.0.D Comdty	PX_LAST	3
rate_expectation	us_rate_expectation_2m	FF2 N:01.0.D Comdty	PX_LAST	3
rate_expectation	australia_rate_expectation_6m	IB6 N:01.0.D Comdty	PX_LAST	3
rate_expectation	australia_rate_expectation_2m	IB2 N:01.0.D Comdty	PX_LAST	3

7 Feature Engineering

We start off with a total of 206 features to start with. Next, we conduct feature engineering on these variables by applying several transformations and also create new features. Finally, we come up with a total of 501 features approx. (including 87 of the initial features taken as is (i.e. features like volatility level, surprise indices, market breadth, HY credit spreads etc.)). The next sections describes some of the feature engineering methods in details,

7.1 Initial transformation by code

I apply few transformations on the main set of variables derived from BBG. This is done so that I can generate more meaningful features afterwards. The transformations are tagged based on a code, ranging from 1 to 4.

- TCode: 1 - % Change
- TCode: 2 - Log Difference
- TCode: 3 - No Transformation
- TCode: 4 - First Difference/100

Next I use the new transformed dataframe to generate relevant features for our models.

7.2 Mahalanobis Distance and Absorption Ratio

Following Kritzman et. al, we use the sector level total return indices gathered from DataStream to create two indicators.

- The first measure is called Financial Turbulence, which is a measure to capture conditions when asset prices move in uncharacteristic fashion, likely due to extreme price moves, decoupling of correlated assets and convergence of uncorrelated assets. The indicator d_t is measured as,

$$d_t = \frac{1}{N} (x_t - \mu) \Sigma^{-1} (x_t - \mu)' \quad (12)$$

x_t is vector of sector returns at time t , μ is the conditional mean upto t and Σ is the covariance matrix of sector returns, also calculated on an expanding window. We compute this measure for each of the developed market indices, using their sector level return series.

- The second measure we measure is called the absorption ratio, which is a measure to capture systemic risk. The goal of this indicator is to capture the relative importance of each sector's contribution to systemic risk, which may not be captured by naive correlation based measures. Following Kritzman et al., we define Absorption Ratio as the fraction of total variance explained by a finite set of eigenvectors. the Absorption Ratio, denoted as AR_t is defined as,

$$AR_t = \frac{\sum_{i=1}^K \sigma_i^2}{\sum_{i=1}^N \sigma_i^2} \quad (13)$$

$$Z_{AR,t} = \frac{AR_t - AR_{252D,t}}{\sigma_{AR-252D,t}} \quad (14)$$

σ_i^2 is simply the eigenvalue of the i th eigenvector, with total of N eigenvectors and K being the first one fifth of them. The eigenvalues were extracted using a rolling correlation matrix of the index sectors with 2 year lookback period. The final indicator was obtained by simply by transforming the raw absorption ratio into z score with 252 day lookback.

7.3 Risk Adjusted Momentum, Equity Skewness, and Kurtosis

For the key total return indices (Equities, bonds, FX Fwds, Commodities, and yields, I compute risk adjusted momentum for all the asset classes. I also add skewness and kurtosis measures for equity factor and fixed income constituents. The three different lookbacks, are 22, 66 and 252 days respectively.

For the momentum indicator, the reason for the risk adjustment is to ensure that the magnitude of the indicator is comparable across all asset classes. The indicator, defined as $Z_{RAMOM,t}$ is defined as,

$$Z_{RAMOM,t} = \frac{r_{t-\tau:t}}{\sigma_{t-\tau:t}} \quad (15)$$

$r_{t-\tau:t}$ is the cumulative return for the given lookback period τ and $\sigma_{t-\tau:t}$ is the τ day volatility calculated over the same period.

The equity skewness and kurtosis measures are simply based on the textbook definitions,

$$Skewness_t = \frac{\sum_{i=1}^N (X_i - \mu)^3}{(N - 1) * \sigma^3} \quad (16)$$

$$Kurtosis_t = \frac{\sum_{i=1}^N (X_i - \mu)^4}{(N - 1) * \sigma^4} \quad (17)$$

7.4 US Total Option Volume

The daily US total option volume for calls and puts are very noisy, so I simply create two features with moving average crossover overlay with 22, 66 day pair and 22, 252 day pair. The new features, denoted as $USOPTVOL_{mac,t}$ can be defined as,

$$MACUSOPTVOL_t = MAUSOPTVOL_{22D,t} - MAUSOPTVOL_{\tau,t-1} \quad (18)$$

$MAUSOPTVOL_{\tau,t}$ is the moving average of the time series with τ lookback period.

7.5 Correlation Between Key Assets

I also include cross asset correlation based features, with equity and fixed income indices being the common part of the pair. The two other asset classes are the corresponding fixed income indices and FX Forwards for a given country. The calculation

is done on a rolling window with 126 day lookback period.

7.6 ATM Implied Volatility Slopes

I also construct several features based on the term structure of implied volatility surface, notably for SPX Index, several USD Swaptions and also FX Options pairs. The general formulation is as follows,

I define the slope signal, denoted as $IVSLOPE_{\tau_1|\tau_2,t}$ for any two maturities, τ_1 and τ_2 as,

$$IVSLOPE_{\tau_1|\tau_2,t} = \log\left(\frac{IVOL_{\tau_1,t}}{IVOL_{\tau_2,t}}\right) \quad (19)$$

$IVOL_{\tau_1,t}$ is the τ_1 implied volatility at time t . Next, the final feature is simply a breakout signal of $IVSLOPE_{\tau_1|\tau_2,t}$. Next, I smoothed the indicators with an EWMA, 21 day span.

7.7 Positioning Data

I also create couple of features using the CFTC positioning data, which can potentially act as a measure of market crowdedness. The crowdedness features are created using positioning data on SPX Index, NDX Index, US2y Treasury, US10y Treasury and Oil. I measure crowdedness. defined as C_t as,

$$C_t = \frac{NetPosition_t}{OpenInterest_t} \quad (20)$$

Next, I smooth the indicator with a simple breakout transformation.

$$BOC_t = C_t - MAC_{52W,t} \quad (21)$$

$MAC_{52W,t}$ is the 52 week moving average of C_t . One thing to note that the positioning data is released weekly every Tuesday. Thus, we construct the indicator using weekly data and then merge it back into the daily dataframe.

7.8 VIX Term Structure Information

Using the VIX Futures Term Structure, I generate the following variables, - **Roll Yield:**

The VIX roll yield (RY_t) is computed as,

$$RY_t = \frac{UX1_t}{VIX_t} \quad (22)$$

- **VIXF slopes at different points of the curve:**

These variables are simply different combination of ratios between the different points of VIXF term structure, to capture all the potential variations not captured by the primary slope measure. The several combinations are as follows,

$$\lambda_{VIXF1:VIXF2,t} = \frac{UX2_t}{UX1_t} \quad (23)$$

$$\lambda_{VIXF1:VIXF3,t} = \frac{UX3_t}{UX1_t} \quad (24)$$

$$\lambda_{VIXF4:VIXF7,t} = \frac{UX7_t}{UX4_t} \quad (25)$$

$$\lambda_{VIXF1:VIXF9,t} = \frac{UX9_t}{UX1_t} \quad (26)$$

- **VIX slopes at different points of the curve:**

Same as the futures, I compute several slope measures, this time using the Cash VIX term structure. The several combinations are as follows,

$$\lambda_{VIX1:VIX3,t} = \frac{VIX3M_t}{VIX_t} \quad (27)$$

$$\lambda_{VIX1:VIX6,t} = \frac{VIX6M_t}{VIX_t} \quad (28)$$

$$\lambda_{VIX1:VIX12,t} = \frac{VIX1Y_t}{VIX_t} \quad (29)$$

$$\lambda_{VIX3:VIX6,t} = \frac{VIX6M_t}{VIX3M_t} \quad (30)$$

- Put/Call:

I also incorporate one volume based information, the put-call ratio of VIX options. Unlike the traditional put-call ratios, which incorporate volume based information. I use open interest to compute the ratio. Using open interest instead of volume puts the focus only on active contracts.

7.9 Term Spreads

Next, I also compute term spreads using the nominal yields for the selected DMs. The term spread indicator, denoted as $TS_{\tau,t}$ is calculated as,

$$TS_{\tau,t} = y_{\tau,t} - y_{3m,t} \quad (31)$$

$y_{\tau,t}$ is either 3 year or 10 year nominal yield for a country and $y_{3m,t}$ is the risk free rate.

7.10 Implied Volatility Skews

I also construct several skew measures using the volatility surface data. For a given moneyness (K/S), I compute $SKEW_{i,t}$ as,

$$SKEW_{i,t} = \frac{IVOL_{m_{90\%},t} - IVOL_{m_{110\%},t}}{IVOL_{ATM,t}}$$

Next, I smoothed the indicators with an EWMA, 21 day span.

7.11 Variance Distance

Variance distance ratio is simply the difference between long term and short term realized variance. This measure, usually computed with returns, I adopt it with for realised volatility to quantify whether current market volatility is abnormal relative to some longer lookback window. I compute $d_{\sigma^2,t}$ for the equity market indices.

$$d_{\sigma^2,t} = \sigma_{22D,t}^2 - \sigma_{252D,t}^2 \quad (32)$$

7.12 Gold-Silver Ratio

I take the 252day return differential between gold and silver as a measure for risk off regimes.

After the aforementioned transformations, we come up with a total of 501 features approx. including the features which are kept unchanged from initial data pull (i.e. features like volatility level, put implied volatility for major indices, economic and inflation surprise indices, market breadth indices, earnings revisions, credit spreads etc.). We merge all of them in one dataframe to create a final dataset.

8 Feature Selection

Given the large number of features constructed, I apply a simple univariate feature selection rule before feeding them in the machine learning model. I use two non-linear approaches to feature selection,

- Point Biserial Correlation: A correlation measure between continuous and binary variables.

$$\rho_{PB} = \frac{\mu_1 - \mu_0}{\sigma_x} \frac{\sqrt{n_1 n_0}}{n} \quad (33)$$

μ_1 and μ_0 are the means of the continuous variables sorted by the binary outcomes. n_1 and n_0 are number of observations in each outcome. σ_x is the standard deviation of the continuous variable.

- Mutual Information: An entropy based measure of codependence between two distributions. This is a nonparametric method based on entropy estimation from k-nearest neighbors distances.

$$I(X, Y) = H(X) + H(Y) - H(X, Y) \quad (34)$$

$H(\cdot)$ is the information measure, in this case entropy.

I compute these metrics for each features, rank them and only pick a subset from each based on a 50th percentile threshold. The final set of features being fed in the model is simply the intersection of the two lists.

The feature selection process is conducted as an initial filter for every recursive estimation done through our sample period.

9 Candidate Models & Estimation Methodology

Accompanied with the features selected from the process described above, I estimate a range of supervised classification models for predicting crisis regimes,

- **Logistic Regression**
- **Support Vector Classifier**
- **Extreme Gradient Boosting**
- **Random Forest**
- **Neural Network**

As a benchmark, I also fit a probit model with 5 principal components from the set of 430 features at each estimation period. The probit model is a very well known methodology widely used in the industry for recession modelling and pca is a common dimension reduction technique used to incorporate information of large data sets.

In order to ensure good out of sample predictability, it is crucial to tune the hyper parameters for each of these models through a cross validation process. Given that our data is time series, I conduct a 4 fold temporal cross validation for each in sample period. I use random search for the cross validated parameter tuning for any model with more than 2 hyper parameters, grid search otherwise. The figure below illustrates the train-test validation process for one iteration. At time t , which is the calibration date, we take into account all information from the start of our dataset t_0 and calibrate all 6 candidate models (including the probit benchmark). For the cross validation procedure used for hyper parameter tuning, we create four splits of the training data. Where for each split, the blue area corresponds to the cross validation train and the green area corresponds to the cross validation test set. The models are trained at each of these blue areas and their performance is evaluated at each of the green blocks. The optimal set of hyperparameters are the ones with the highest average accuracy across all cross validation test set. For models with greater than two hyper parameters, we apply a random search procedure and grid search otherwise.

At time t , we incorporate all the information from inception. Next, we apply the feature selection process on the training data to come up with a short list of features. Next, we scale all the selected features between -1 and 1. Next, we conduct 4-fold time series cross validation for each of the models and come up with a set of optimal hyper parameters. Using the

Figure 3: Illustration of Train-Cross Validation-Test Splits for One Iteration

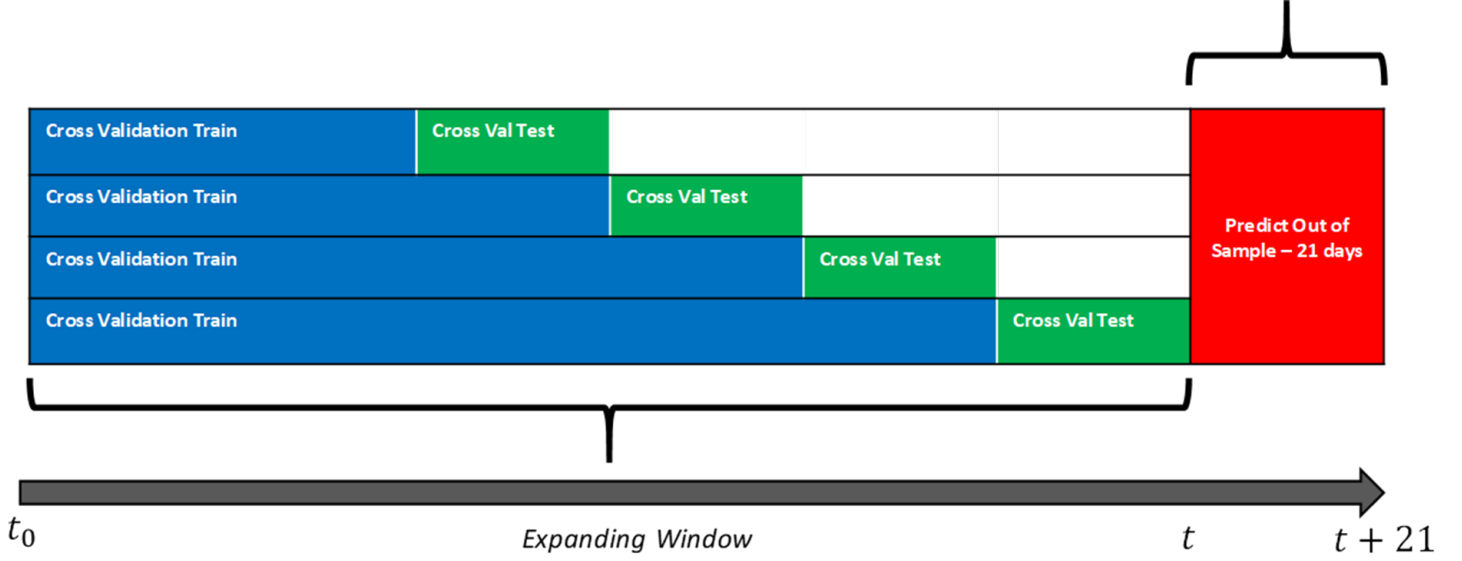
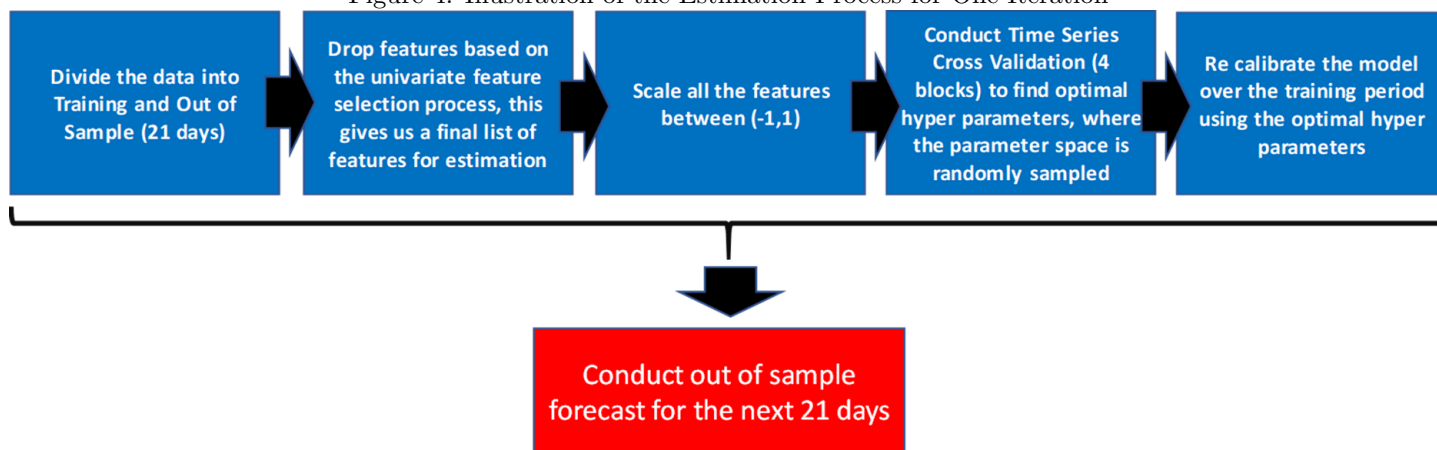


Table 6: Hyperparameter Space for All Classifiers

Models Hyperparameters	Logistic Regression	Support Vector Classifier	XG Boost	Random Forest	Neural Network
alpha					[1.00e-04 1.58e-03 2.51e-02 3.98e-01 6.31e+00 1.00e+02]
C	[1.e-03 1.e-02 1.e-01 1.e+00 1.e+01 1.e+02]	[1.e-03 1.e-02 1.e-01 1.e+00 1.e+01 1.e+02]			
max_depth			[3, 5, 9, 11, 13, 15, 20]	[3, 5, 9, 11, 13, 15]	
min_child_weight			[0.5, 1.0, 3.0, 5.0, 10, 20]		
subsample			[0.3, 0.5, 0.7, 0.9]		
colsample_bytree			[0.3, 0.5, 0.7, 0.9]		
learning_rate			[0.01, 0.1, 0.2, 0.3, 0.5]		
n_estimators			[100, 500, 1000]	[100, 200, 500, 750, 1000]	
lambda			[0.001, 0.01, 0.3, 0.5, 1, 3, 5, 7]		
min_samples_leaf				[0.001, 0.003, 0.005, 0.01, 0.025, 0.05]	
min_samples_split				[0.001, 0.003, 0.005, 0.01, 0.025, 0.05]	
max_features				[0.05, 0.1, 0.3, 0.5]	
hidden_layer_sizes					[(32, 16, 8), (32, 16), (32,)]

optimal hyper parameters, we re-calibrate the model in the training period and use the fitted model to generate out of sample predictions.

Figure 4: Illustration of the Estimation Process for One Iteration



For each recursive estimation, I predict out of sample probabilities for the next one month, i.e. 21 days. The diagram below shows a general illustration of the estimation process.

10 Performance Evaluation

10.1 Overview of Key Metrics

Below, I briefly describe some of the key performance metrics used to evaluate the models.

ROC Area Under the Curve: An ROC curve for is a graph which shows the performance of a classification model for all possible probability thresholds. The curve is generated by plotting the true positive rate against the false positive rate for all possible thresholds. The area under the curve provides an aggregate measure across all possible thresholds. One way of interpreting AUC is as the probability that the model ranks a random positive example more highly than a random negative example. The AUC of a random model is 0.50 and the highest AUC can be achieved is close to one.

Average Precision Score: Average precision score summarizes a precision recall curve as a weighted mean of precision achieved at each threshold, with the delta in recall used as weights.

Brier Skill Score: Brier Score is simply the mean squared error between the probability forecast p_t and the actual outcome y_t .

$$BS = \frac{1}{N} \sum_{t=1}^N (y_t - p_t)^2 \quad (35)$$

The Skill Score measure how good our forecast is compared to some reference benchmark.

$$BSS = 1 - \frac{\sum_{t=1}^N (y_t - p_t)^2}{\sum_{t=1}^N (\hat{p}_t - p_t)^2} \quad (36)$$

Here, the reference or the benchmark is simply the insample average or denoted as $\hat{p}_t$ up until the beginning of the out of sample forecast.

- **Accuracy:** Accuracy for our out of sample forecast is calculated as,

$$Accuracy = \frac{TPR + TNR}{2} \quad (37)$$

F1 Score: This measure is calculated based on the harmonic mean of precision and recall. This is a good measure to use in cases when our data is imbalanced and we would like balance between lower False Positive and False Negatives.

$$F1 = \frac{2}{Recall^{-1} + Precision^{-1}} \quad (38)$$

Matthews Correlation: MCC is calculated as,

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (39)$$

This measure ranges between -1 and 1 and is very robust against imbalanced data.

10.2 Analysis of Evaluation Metrics

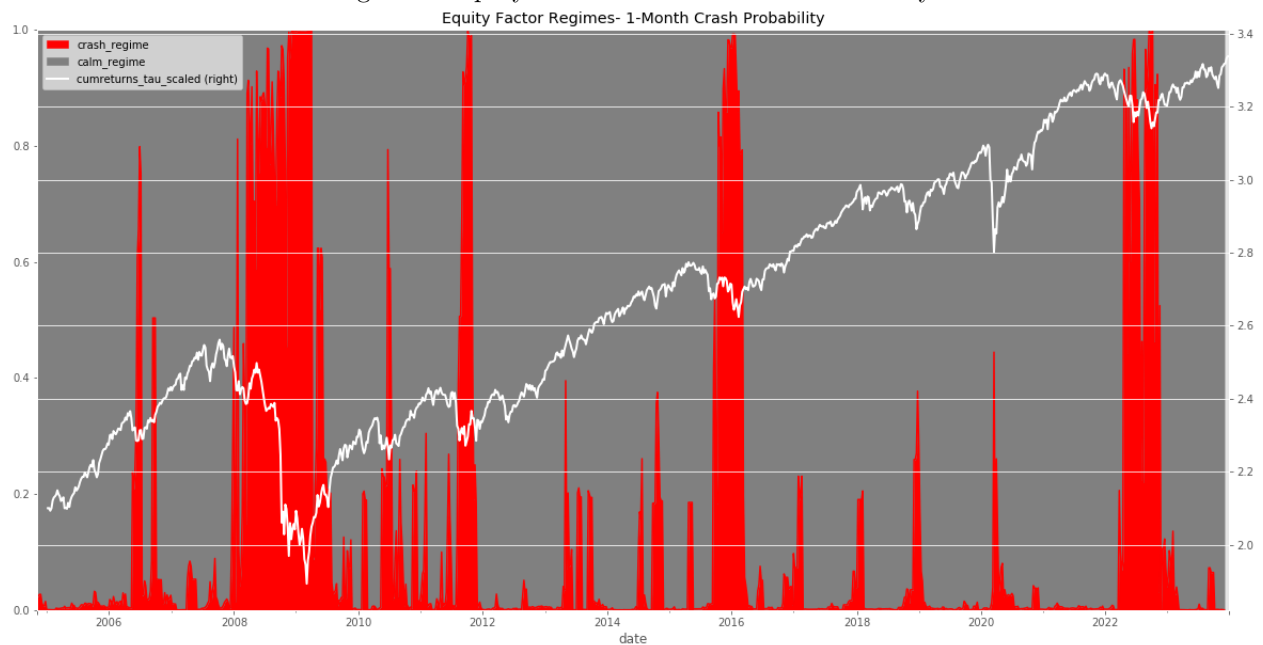
Below, we present the evaluation metrics for all the candidate models. The Logit model ranks the lowest across all metrics. XGBoost and Neural Network rank tie as the top models, with higher scores across all metrics. However, the ensemble predictions, which is based on the mean of all the candidate models ranks as the best. The simple Probit PCA model outperforms logistic regression and ties with support vector machine.

Figure 5: Performance Evaluation - Classification Metrics

	Probit PCA	Logistic	Support Vector Machine	Random Forest	Neural Network	XGBoost	Ensemble
AUC	0.77	0.78	0.79	0.82	0.84	0.81	0.81
Average Precision Score	0.48	0.61	0.45	0.50	0.54	0.47	0.52
Brier Skill Score	0.25	0.35	0.36	0.42	0.40	0.39	0.47
Precision	0.48	0.20	0.60	0.37	0.30	0.43	0.43
Recall	0.51	0.79	0.62	0.77	0.81	0.68	0.74
F1-Score	0.49	0.32	0.61	0.50	0.44	0.53	0.54
Accuracy	0.70	0.62	0.77	0.77	0.74	0.76	0.78
Matthews Correlation	0.40	0.17	0.54	0.41	0.35	0.44	0.46
Rank	5.50	7.00	5.50	4.00	2.50	2.50	1.00

The figure below plots the predicted probabilities from XG Boost (smoothed with a 21 day EWMA), on top of the Equity Factor.

Figure 6: Equity Factor 1-Month Crash Probability



11 Position Sizing using Regime Probabilities

Next, in order to check whether the models incorporate ex ante forecast which yield information useful for portfolio allocation purposes, I run a simple trading strategy on the equity factor using predicted probabilities. Under the null hypothesis $H_0 : p[y_t = 1] = \alpha$, with α being the random chance. (we use $\alpha = 0.50$ for illustration), we compute the test statistic $Z_{timing,t+1}$ as,

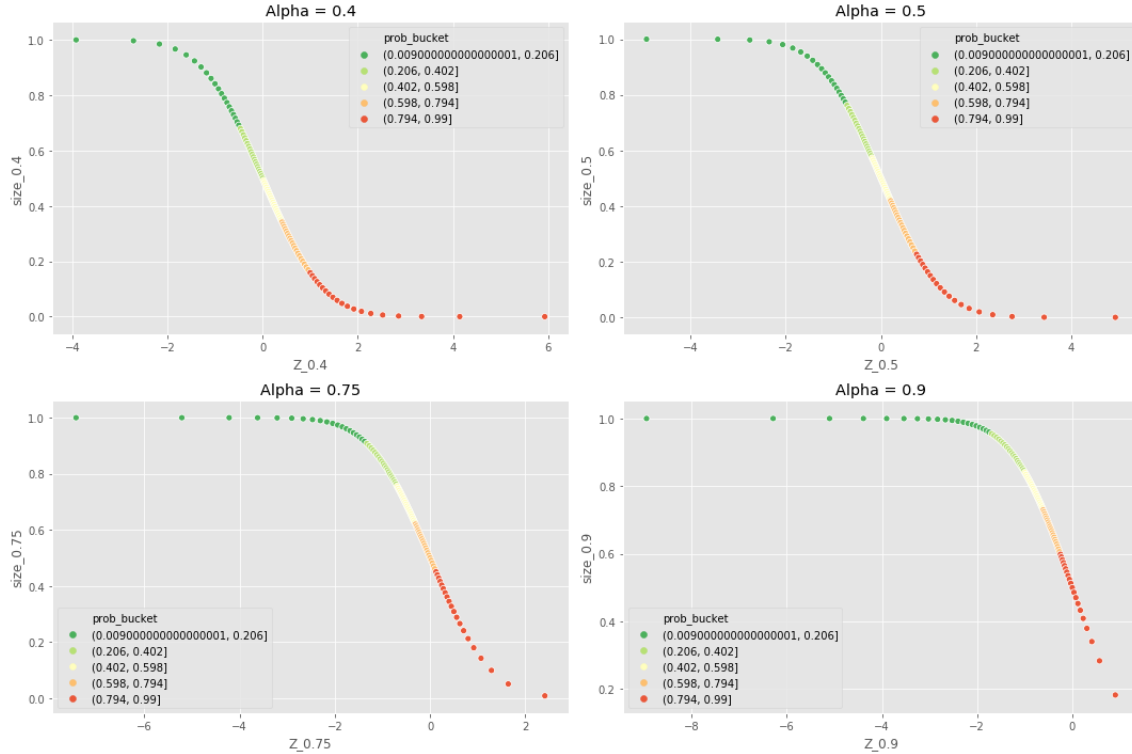
$$Z_{timing,t+1} = \frac{p_t - \alpha}{\sqrt{p_t(1 - p_t)}} \quad (40)$$

The position size, denoted as $X_{timing,t+1}$ can be written as,

$$X_{timing,t+1} = \Phi(-Z_{timing,t+1}) ; X_{timing,t+1} \in [0, 1] \quad (41)$$

The next figure shows a simple illustration of the formula, assuming for four different values for α . As we can see that, position sizes can vary dramatically based on what the investors' view of random chance is. A lower value of α will denote lower tolerance to taking more risk. In such cases one would potentially scale down their exposure a lot faster relative to a case where α is very high, which suggest lower level of risk aversion. Next, we can use the position sizes obtained from all six models to conduct a simple backtest on the equity factor, with weekly rebalancing.

Figure 7: Performance Evaluation - Position Size under Different α



As show in the next table, the results from the simple backtests are consistent with the performance metrics shown in the previous sections. XG Boost is the best performing strategy with a 0.80 sharpe ratio with SVC strategy ranking as the second best. Also, the probit model has a sharpe ratio of 0.71, still slightly higher than random forest (0.69). However, all the models demonstrate better historical performance compared to a long only equity factor strategy, which has a sharpe ratio of 0.52 for the back test period (2005 till present).

Figure 8: Performance Evaluation - Position Sized based on Probabilities

	Probit PCA	Logistic	Support Vector Machine	Random Forest	Neural Network	XGBoost	Ensemble
Arith. Mean	8.42%	7.16%	9.49%	7.54%	8.15%	9.33%	8.04%
Compound Mean	8.00%	6.90%	9.11%	7.17%	7.84%	9.00%	7.74%
Volatility	11.92%	9.72%	12.21%	10.87%	10.83%	11.71%	10.68%
SR	0.71	0.74	0.78	0.69	0.75	0.80	0.75
t-Stat	3.02	3.15	3.32	2.96	3.22	3.40	3.22
Geo SR	0.67	0.71	0.75	0.66	0.72	0.77	0.72
Max DD	25.08%	19.10%	30.54%	26.80%	26.80%	30.34%	26.08%
Ulcer	1.06	1.37	1.49	0.94	1.35	1.30	1.27
Calmar	0.34	0.38	0.31	0.28	0.30	0.31	0.31
Skewness	-1.09	-1.12	-1.09	-1.10	-0.92	-1.29	-1.17
Kurtosis	6.34	8.52	7.41	6.55	9.25	8.26	7.29

12 Performance Benchmark Analysis

A few question can raise in the readers mind related to our model, such as "how different the ML model implied performance from using a simple markov switching model, or simply running a simple trend following strategy or a timing signal derived from VIX?". To answer the question, I benchmark my ensemble forecasts from the machine learning models against four different benchmarks, with two different strategies for each, long-out (LO) and long-short (LS). All the strategies are rebalanced weekly.

12.1 Overview of Strategies

Simple Trend Following: The simple trend following rule is based on the average breakout calculated based on three different lookbacks.

$$X_{strat,t+1}^{Trend} = \begin{cases} 1; & \text{if } MABO_t > 0 \\ -1; & \text{if } strat = LS \text{ \& } MABO_t < 0 \\ 0; & \text{if } strat = LO \text{ \& } MABO_t < 0 \end{cases} \quad (42)$$

$$MABO_t = \frac{1}{3} \sum_{lkbk \in [22, 63, 261]} (EQ_t - EQMA_{t-1, lkbk}) \quad (43)$$

EQ_t is the Equity Factor at t , and $EQMA_{t, lkbk}$ is the moving average of the equity factor for a given lookback.

VIX Timing Signal: The VIX based timing signal is calculated as the ratio between the level of VIX at t and the level of VIX at a given percentile calculated based on an expanding window, to incorporate full historical information at a given point in time. The timing signal is calculated as,

$$X_{strat,t+1}^{VIX} = \begin{cases} 1; & \text{if } VIXR_t > 0 \\ -1; & \text{if } strat = LS \text{ \& } VIXR_t < 0 \\ 0; & \text{if } strat = LO \text{ \& } VIXR_t < 0 \end{cases} \quad (44)$$

$$VIXR_t = \frac{VIX_t}{VIX_{t-1,\alpha}} \quad (45)$$

$$VIXRBO_t = \frac{1}{3} \sum_{lkbk \in [22,63,261]} (VIXR_t - VIXRMA_{t-1,lkbk}) \quad (46)$$

Here, $VIX_{t-1,\alpha}$ is the 95th percentile over an expanding window and $VIXRMA_t$ is moving average of $VIXR$ for a given lookback.

Volatility Targeting: The 10% volatility targeting strategy is simply based on the leverage ratio, which is calculated as

$$X_{t+1}^{VolTgt} = \frac{10\%}{\sigma_{ewma,t}} \quad (47)$$

$\sigma_{ewma,t}$ is calculated based on a conditional covariance matrix constructed from an exponentially weighted correlation matrix with span of 261 days and exponentially weighted volatility for each constituent with a span of 63 days.

Hidden Markov Model (HMM): For the HMM model, the state prediction derived from HMM is based on the probability estimates. The model is based on two states (similar to our ML model) and calibrated on the Financial Turbulence measure (following Kritzman et. Al). After fitting HMM on the underlying data over an expanding window, we can derive the conditional regime probabilities and the corresponding predicted states.

XG Boost Crash Probability: The Crash Model Signal is defined based on a cut off threshold of 0.50,

$$X_{strat,t+1}^{ML} = \begin{cases} 1; & \text{if } X_{timing,t} > 0.50 \\ -1; & \text{if strat = LS \& } X_{timing,t} < 0.50 \\ 0; & \text{if strat = LO \& } X_{timing,t} < 0.50 \end{cases} \quad (48)$$

12.2 Performance Comparison & Spanning Tests

In this section, I compare and contrast the performance of the benchmarks. The table below outlines performance statistics of the backtests. The long only equity factor has a sharpe ratio of 0.58. The closest comparison to the equity factor will be the long-out strategies. Most long-out approaches provides a higher risk adjusted return compared to the equity factor, with ML ensemble having the highest realized sharpe ratio of 0.85. The Trend and VIX based strategies performed similar for the sample period, with a long-out sharpe ratio of 0.76 and 0.75. Next, looking at the long-short strategies, the machine learning model is again the best performer, with a realized sharpe ratio of 0.78. The VIX based long-short also performs well with a sharpe ratio of 0.66. The trend and hmm strategies are the worst long-short performers, suggesting higher false positives in the predicted signal compared to the VIX based approach and Machine Learning Ensemble.

Figure 9: Benchmark Analysis - Performance Results

	Arith. Mean	Compound Mean	Volatility	SR	t-Stat	Geo SR	Max DD	Ulcer	Calmar	Skewness	Kurtosis
Equity Factor	9.25%	8.26%	16.0%	0.58	2.52	0.52	53.2%	0.71	0.17	-0.99	8.94
trend_strat_lo	7.35%	7.11%	9.7%	0.76	3.30	0.73	16.7%	1.33	0.44	-0.86	6.21
vix_strat_lo	9.92%	9.45%	13.3%	0.75	3.26	0.71	27.4%	1.26	0.36	-0.53	2.50
hmm_turbulence_strat_lo	6.63%	6.08%	11.9%	0.56	2.42	0.51	30.6%	0.50	0.22	-1.23	8.34
hmm_vol_strat_lo	3.68%	3.14%	10.7%	0.34	1.50	0.29	40.9%	0.19	0.09	-1.68	12.16
voltgt_strat_lo	7.88%	7.38%	12.2%	0.65	2.82	0.61	31.0%	0.73	0.25	-1.25	5.37
crashprob_strat_lo	10.85%	10.55%	12.7%	0.85	3.73	0.83	30.6%	1.66	0.35	-0.76	7.18
trend_strat_ls	5.45%	4.25%	16.1%	0.34	1.48	0.26	27.5%	0.50	0.20	0.36	8.48
vix_strat_ls	10.59%	9.75%	16.0%	0.66	2.89	0.61	33.3%	1.17	0.32	0.23	8.57
hmm_turbulence_strat_ls	4.00%	2.75%	16.1%	0.25	1.09	0.17	60.2%	0.14	0.07	-0.17	8.56
hmm_vol_strat_ls	-1.89%	-3.14%	16.1%	-0.12	-0.51	-0.20	73.9%	-0.04	-0.03	-0.12	8.51
crashprob_strat_ls	12.45%	11.81%	16.0%	0.78	3.40	0.74	30.6%	1.62	0.41	0.03	8.70

A more formal performance analysis to assess the performance of ML model is to conduct a spanning tests with respect to the benchmark strategies. Using the weekly returns, I run two different regressions, one for long-out and the other for

long-short performance,

$$r_{MLCrashlo,t} = \alpha_{lo} + \beta_{trendlo}r_{trendlo,t} + \beta_{VIXlo}r_{VIXlo,t} + \beta_{hmmlo}r_{hmmlo,t} + \beta_{VolTgt}r_{VolTgt,t} + \epsilon_{lo,t} \quad (49)$$

$$r_{MLCrashls,t} = \alpha_{ls} + \beta_{trendls}r_{trendls,t} + \beta_{VIXls}r_{VIXls,t} + \beta_{hmm ls}r_{hmm ls,t} + \epsilon_{ls,t} \quad (50)$$

Figure 10: Benchmark Analysis - Spanning Tests

Spanning Tests	Long/Out			Long/Short		
	Coefs	Return Contribution	t-Stat	Coefs	Return Contribution	t-Stat
alpha ann.		2.87%	2.13		7.28%	2.65
trend_strat	0.258	1.89%	9.01	0.189	1.03%	7.46
vix_strat	0.098	0.97%	3.12	0.215	2.28%	7.87
hmm_turbulence_strat	0.401	2.66%	13.04	0.461	1.84%	17.73
vol_tgt_strat	0.313	2.46%	7.35			
<i>R Squared Adj.</i>	78.90%			44.0%		
<i>Idio. Volatility</i>	5.82%			11.94%		
<i>Idio. Sharpe</i>	0.49			0.61		

The table above shows the results for the spanning tests, In both cases, both simple trend and VIX based strategies have statistically significant exposure to the ML strategy. However, the intercept from both regression is statistically significant at 5% significance level, suggesting that even though some of the total variation in the ML based strategies are explained by the benchmark strategies, they don't fully capture the entirety of the dynamics and leaves out a significant portion of returns as positive alpha. This suggesting that there is information content within the ML based probabilities regarding possible regime shifts ex-ante which simply cannot be captured by the signal generated for the benchmark strategies. The positive alphas under both regressions corresponds to a idiosyncratic sharpe ratio of 0.49 for long-out and 0.61 for the long-short strategy respectively.

Figure 11: Cumulative Performance - Long/Out Strategies

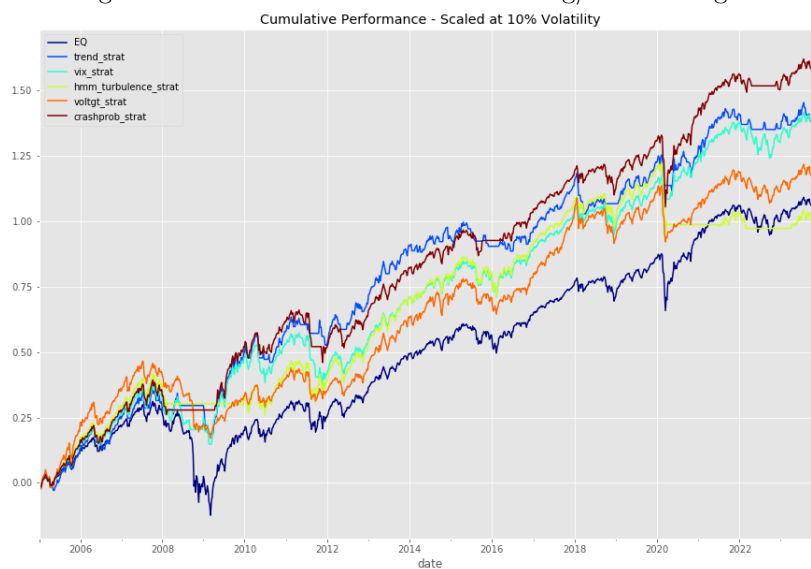
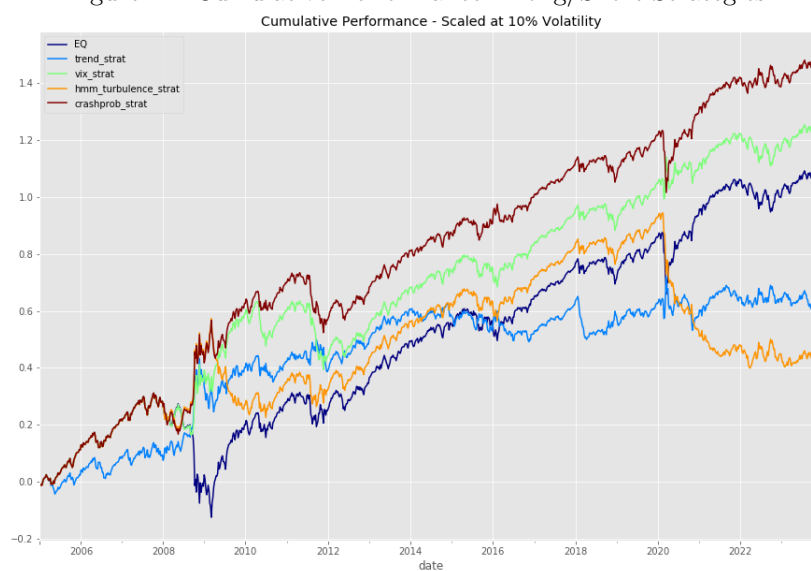


Figure 12: Cumulative Performance - Long/Short Strategies



13 Applications in Portfolio Construction

In this section, I demonstrate two key applications in Portfolio Construction by incorporating the regime probabilities as predicted from XG Boost.

13.1 Rotation between Risk-On & Risk-Off Assets

In the first approach, I show a very simple approach of how to incorporate probabilities as position sizes in a portfolio with discretionary bands. The bands provided can be treated as allocation under different regimes, either risk on or risk off. We start off with a global multi asset portfolio as shown in the table below,

Table 7: Global Multi Asset Portfolio

Asset Name	Assets/Tickers
Global Equities	Equity Factor
Global High Yield	LG30TRUH Index
Energy Commodities	SPGSENTR Index
Non Energy Commodities	SPGSNETR Index
Global Fixed Income	MLT1US10 Index
	SPCACGTR Index
	SPEUBDTR Index
	SPUKGTTR Index
	SPJGBTR Index
	SPAU10DT Index
Global Investment Grade	LEGATRUH Index
Global ILBs	LF94TRUH Index
Gold	SPGCGCTR Index
Multi Asset CTA	SGIXTFXA Index

For global equities, we simply use the equity factor as a proxy. for global fixed income, I use a weighted basket of DM bonds. The table below outlines the portfolios to be analysed. The 'Exposure Bucket' column categorizes the assets in risk-on, risk-off and tail hedge based on their underlying characteristics. The weights defined here are purely based on our discretionary understanding of the assets and doesn't necessarily have to be absolute as per the reader. The 'Baseline' portfolio is our starting point, which comprises a total of 80% in equity centric assets and 20% exposure in assets with interest rate sensitivity. The 'Baseline' portfolio mostly mimicks a typical composition of an institutional portfolio, which tends to be largely equity centric or have substantial exposure to growth risk. The 'Timing Lowerbound' and 'Timing Upperbound' are the lower range and the upper range of our timing portfolio. The general idea is as crash probability increases, portfolio allocation will gradually rotate from the lowerbound portfolio to the upperbound portfolio, based on a position size. One key point to note, the lowerbound weights are quite similar to the baseline portfolio with added exposure to a few alternative assets like gold and multi asset trend following. As for the upperbound weights, the exposure to interest rate sensitive assets increases from 20% to 50%, exposure to global equities stays the same with increased allocation to equity tail hedge to compensate any potential equity losses for not cutting position. I also define two other benchmark portfolios, one with the baseline weights plus some exposure in tail hedge and another portfolio with almost 99% exposure in growth sensitive/ risk on assets plus some tail hedge.

Table 8: Portfolio Weights Under Different Test Cases

	Exposure Bucket	Timing Lower-bound	Timing Upper-bound	Baseline
Global Equities	Risk On	50.0%	50.0%	50.0%
Global High Yield		15.0%	0.0%	15.0%
Energy Commodities		7.5%	0.0%	7.5%
Non Energy Commodities		7.5%	0.0%	7.5%
Global Fixed Income	Risk Off	7.5%	18.8%	7.5%
Global Investment Grade		7.5%	18.8%	7.5%
Global ILBs		5.0%	12.5%	5.0%
Gold		5.0%	8.0%	0.0%
Multi Asset CTA		5.0%	8.0%	0.0%

For the timing portfolio, we allocate to each asset i by scaling between its lowerbound and upperbound,

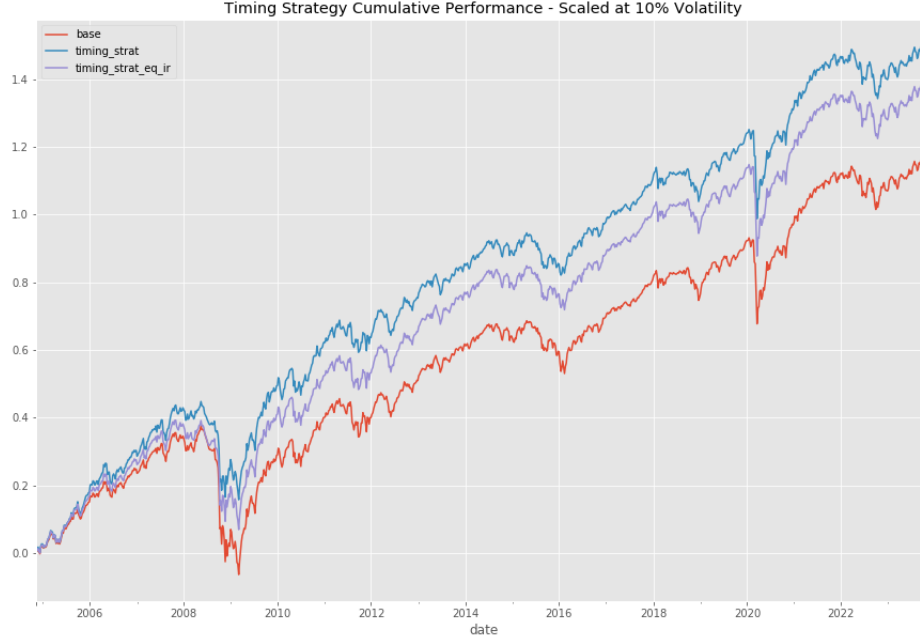
$$w_{i,t} = w_{i,lower} + X_{timing,t+1} * (w_{i,upper} - w_{i,lower}) \quad (51)$$

The table below documents performance statistics for two version of timing strategies, one based on the full set of assets, while another one based on similar assets as the baseline portfolio (EQ + IR). All the portfolios are rebalanced weekly. The two timing strategies produced very promising realized performance, with higher sharpe ratios compared to the baseline portfolio. A proper comparison with the baseline portfolio will be the 'Timing Strategy EQ + IR' portfolio. The performance difference is roughly 0.10 sharpe points, suggesting that even without alternative assets like Gold, and CTA, simply incorporating a dynamic allocation in a traditional equity and interest rate sensitive portfolio yield higher risk adjusted returns. Adding Gold and CTA improved the sharpe ratio, from 0.61 to 0.78.

Figure 13: Timing Strategies - Performance Statistics

	Arith. Mean	Compound Mean	Volatility	SR	t-Stat	Geo SR	Max DD	Ulcer	Calmar	Skewness	Kurtosis
Baseline	6.55%	6.13%	10.80%	0.61	2.65	0.57	39.5%	0.79	0.17	-1.55	12.14
Timing Strategy (EQ + IR)	6.89%	6.64%	9.54%	0.72	3.16	0.70	27.7%	1.21	0.25	-1.32	9.85
Timing Strategy	7.61%	7.39%	9.72%	0.78	3.42	0.76	25.6%	1.40	0.30	-1.20	8.79

Figure 14: Cumulative Performance - Timing Strategies



13.2 Incorporating Regimes in Risk Based Views

Another approach of using the probabilities is to apply them on our views on the assets/asset classes. For example, one can essentially come up with expected return forecast for each regime and then use the probabilities to come up with a single forecast which can be used as an input to our portfolio optimization/construction model. In this case, I took a different approach and show an example where I apply the probabilities for risk forecasting, i.e. forecasting a covariance matrix and apply the covariance matrix in a risk based allocation framework. Since I already have the historical tags for both regimes, I can estimate two covariance matrices, one related to the crash regimes, $\Sigma_{crash,t}$ and the other related to a normal regime $\Sigma_{normal,t}$ (using 1 year rolling lookback). Next, using the out of sample probabilities p_t from XG Boost, I can come with a full covariance matrix as follows,

$$\Sigma_{regime,t} = p_t * \Sigma_{crash,t} + (1 - p_t) * \Sigma_{normal,t} \quad (52)$$

Next, based on a 2-asset portfolio of the Global Equities and Global Fixed Income baskets (as defined in the earlier section), I apply this methodology to 10% Volatility Regime Risk Parity Strategy and compare it to a 10% Volatility naive risk parity, which uses a simple covariance measure (using 1 year rolling lookback). Under the two asset case, the optimal risk parity weight vector at t is,

$$w_{RP,t} = \frac{diag(\Sigma_t)^{-1/2} \mathbf{1}_N}{\mathbf{1}_N' diag(\Sigma_t)^{-1/2} \mathbf{1}_N} \quad (53)$$

Next, for 10% volatility targeting we compute the leverage ratio, λ_t as,

$$\lambda_t = \frac{10\%}{\sqrt{w_{RP,t}' \Sigma_t w_{RP,t}}} \quad (54)$$

Then, the return for both risk parity strategy at t as,

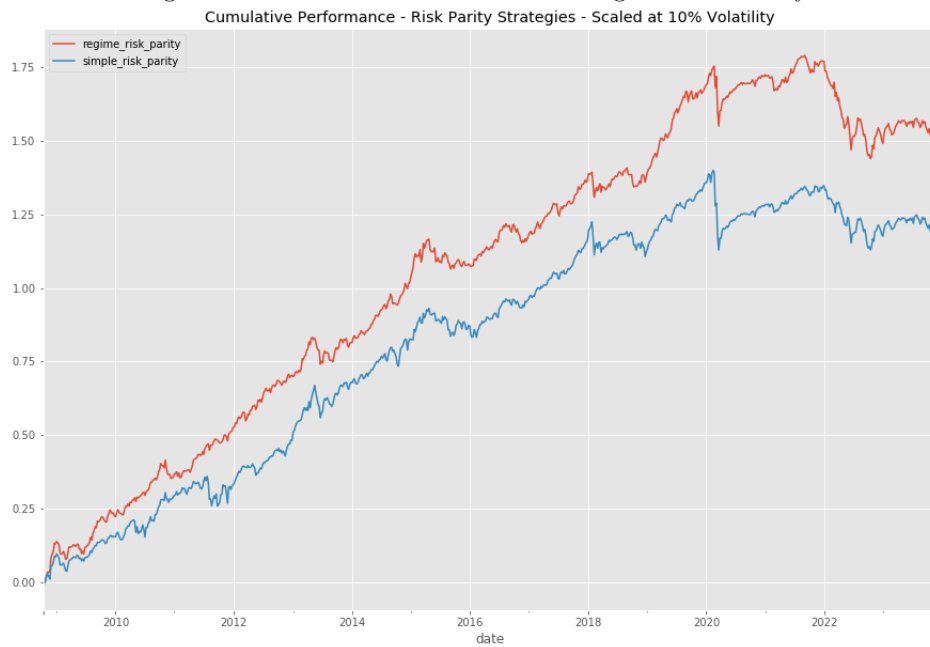
$$r_{RP,t} = \lambda_{t-1} * (w_{RP,t-1}' R_t) \quad (55)$$

R_t is vector of asset returns. The table below presents the performance statistics of both risk parity strategies, the first one based on a covariance matrix as (43) and the latter based on a simple covariance matrix calculated over 1 year rolling window. The regime risk parity has a higher realized sharpe ratio compared to the naive risk parity, thus demonstrating the efficacy of risk forecasting through the incorporation of regime based information.

Figure 15: Performance Statistics - Regime Risk Parity

	Arith. Mean	Compound Mean	Volatility	SR	t-Stat	Geo SR	Max DD	Ulcer	Calmar	Skewness	Kurtosis
regime_risk_parity	11.0%	11.0%	10.5%	1.05	4.07	1.04	31.8%	1.21	0.35	-1.27	8.23
simple_risk_parity	8.1%	7.9%	9.8%	0.83	3.23	0.81	25.1%	0.96	0.32	-1.70	12.23

Figure 16: Cumulative Performance - Regime Risk Parity



14 Interpreting the Predicted Regimes

In this section we tackle the issue of model interpretability in most ML type models. We demonstrate a method called "Shapley Values" to explain our forecasts coming from these machine learning models. Interpreting model prediction is extremely beneficial for both systematic and discretionary investment decisions. In the case of a systematic process where ML based framework is being implemented, an investor needs to understand the underlying feature contribution to gain more confidence about the position sizes of his/her portfolio. On the other hand, in the case of discretionary macro decision making process with Machine learning based framework being as an additional tool, being able to interpret the underlying drivers of an indicator will give the manager a good way to sanity check whether the model output is consistent with their subjective views.

14.1 Introduction to Shapley Values

The interpretation of the Shapley value is: Given the current set of feature values, the contribution of a feature value to the difference between the actual prediction and the mean prediction is the estimated Shapley value. Given the "blackbox" nature of machine learning algorithms, the predictions of the model can be quite unintuitive. Shapley values help tackle with that interpretation issue. The idea was coined from coalitional game theory, where a prediction from a model can be explained by assuming that each feature of an observation to be predicted is a "player" in a game where the "payout" is the prediction. Shapley value tells us how to distribute this "payout" or prediction across the feature.

The best way to understand shapley values is to look at it from a linear modelling framework. A linear model prediction will look sth like,

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_2 x_2 + \beta_j x_j$$

The equation is pretty self-explanatory. Now the contribution, defined as ϕ_j , of the j_{th} variable on the prediction $\hat{y}$ is,

$$\phi_j = \beta_j x_j - E(\beta_j x_j) = \beta_j x_j - \beta_j \mu_j$$

Here, $\beta_j \mu_j$ is the mean effect estimate of j computed using some training data. Shapley values try to quantify ϕ_j for every single feature for a machine learning model output, but using coalitional game theory.

Also, if we sum all the ϕ_j across all the features of an input and add back the mean effect estimate for each variable, we get back the original prediction. This means shapley values are also additive across features, for a given prediction. That is,

$$\sum_{j=1}^p \phi_j = \sum_{j=1}^p (\beta_j x_j - \beta_j \mu_j) = \hat{y} - E(f(x))$$

Shapley values does a similar local decomposition for a machine learning model, using a simulation based approach.

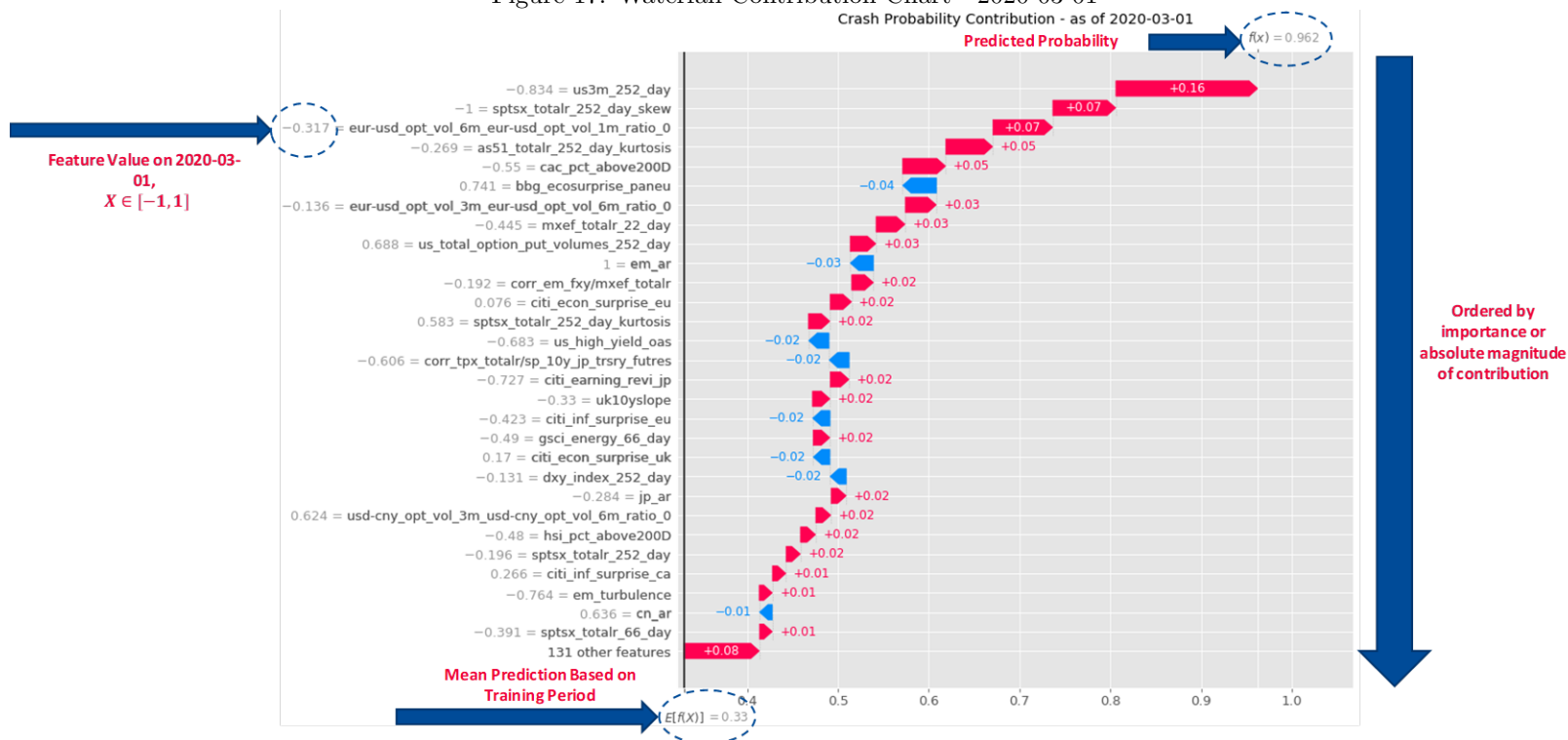
14.2 Shapley Values - An Hypothetical Example with COVID Crash

In this section, we illustrate an example with shapley values. We analyse the predicted probability for the March 1st 2020, which is approximately 2 weeks prior to market sell off less than our defined threshold. The figure below is a waterfall chart where the y-axis lists the features ordered by its importance or based on the absolute magnitude of the contribution. The circled $f(x)$ on the upper right corner is the predicted probability for that particular date, which is 96.2%. On the bottom left, $E[f(x)]$ is the average prediction over the training period prior to March 1st. Each value on the plot explains the difference between the actual prediction (96.2%) and the average prediction (33%) by the corresponding feature, while adding up all the values will equate to the total difference (which is 63.2%).

From the water fall chart, we see the top most feature which contributed the strongest in explaining the deviation from the average prediction is the US 3 month risk free rate momentum, which is a possible indication of flight to quality. We also see a few feature related to realized higher order moments (like skewness and kurtosis), which suggests information about

the tail of the distribution being incorporated in the prediction. Also, a majority of the top features for that date is related to the Euro Area (i.e. EURUSD Implied Volatility Slope, CAC Breadth, and, EU Economic Surprise etc.). This is intuitive given that during the last week of February 2020 and early March 2020, several regions in the Euro Area were grappled with COVID 19 spread, which also created panic across markets.

Figure 17: Waterfall Contribution Chart - 2020-03-01



Next, we can also inspect the dependence plots for the top features to take deeper dive into the relationship between the feature contribution and the feature itself. The figure below shows dependence plots for the six top features. The y-axis denotes the shapley values or the prediction contribution of a feature and x-axis is the feature itself. For example, the dependence plot for 'sptsx_totalr_252_day_skew' demonstrates a non-linear relationship, with probability prediction increasing at a faster rate as the distribution of returns becomes more negatively skewed. Furthermore, let's also inspect the dependence plot for 'eur-usd_opt_vol_6m-usd_opt_vol_1m_ratio_0'. Ideally a upward sloping implied volatility term structure is expected during normal times, however during market turbulence we would expect the slope to be inverted. That's exactly the plot show, more negative/inverted slope corresponds to a higher rate of increase in crash probability.

Figure 18: Feature Dependence Plots

